

Bar Profiles Code Flow Overview

High-level map of the inputs, scripts, helper functions, and generated outputs.

The repository now has three related workflows: image download and geometry linking, isophote/profile plotting, and paper-figure reproduction from catalogue tables.



Key distinction

- Image plots are derived from FITS images plus geometry.
- Figures 6, 9, 10, and 11 are reproduced from catalogue tables and human classifications.
- PE and VPD profile classes are not calculated by the code; they are read from classification files.

Output PDF: [barprofiles_code_flow_overview.pdf](#)

Practical Reading Order

Open these Python files first; then drill into the functions listed inside them.

1. Download images

Open: `s4g_image_downloader/download_s4g_images.py`

Read: `main()` -> `read_galaxy_names()` -> `download_image()`

2. Build the geometry manifest

Open: `s4g_image_downloader/build_s4g_geometry_manifest.py`

Read: `main()` -> `build_manifest_rows()` -> `read_s4g_table()` -> `find_image_file()`

3. Draw isophote/profile plots

Open: `s4g_image_downloader/plot_s4g_isophote_axes.py`

Read: `main()` -> `make_plot()` -> `profile_at_pa()` -> `deprojected_profile_radius()`

4. Reproduce Figure 6

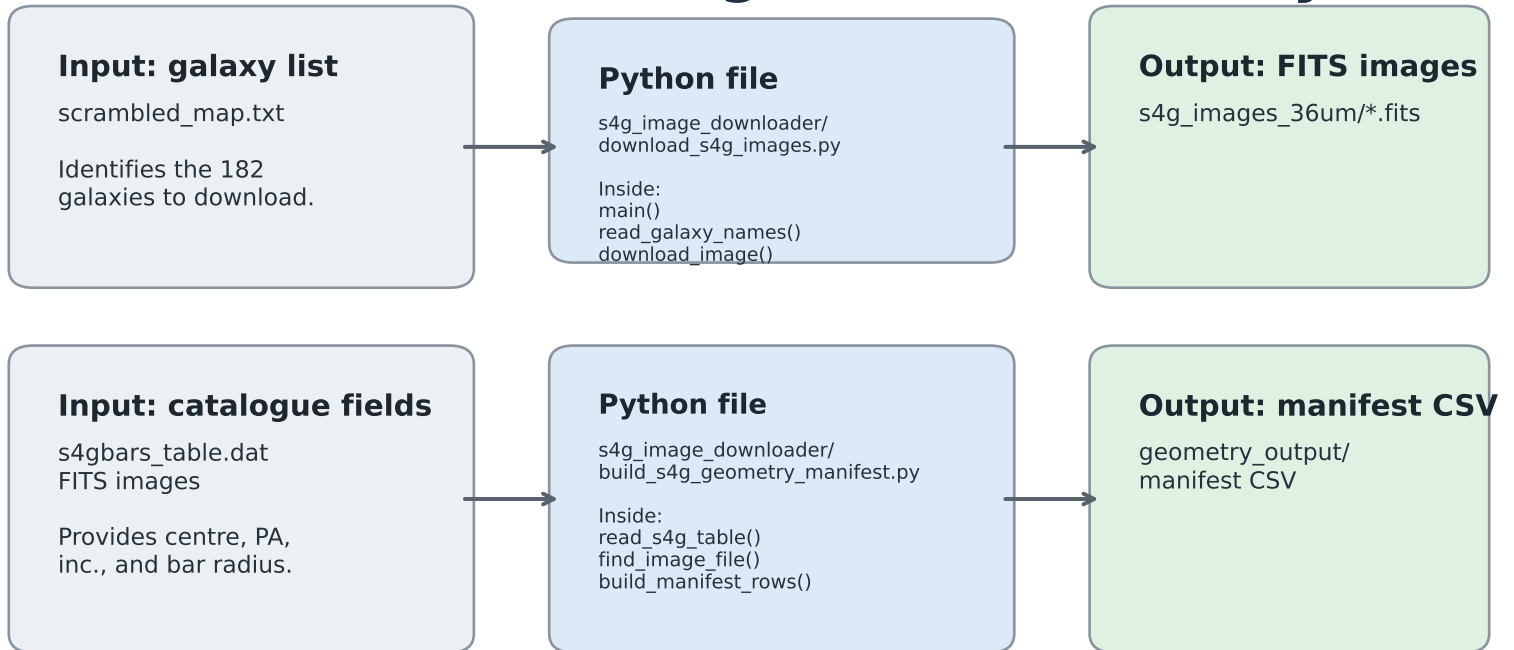
Open:
`barprofiles_paper_GB_working_copy/
reproduce_figure6_histograms.py`

5. Reproduce Figures 9-11

Open:
`barprofiles_paper_GB_working_copy/
reproduce_figures9_10_11.py`

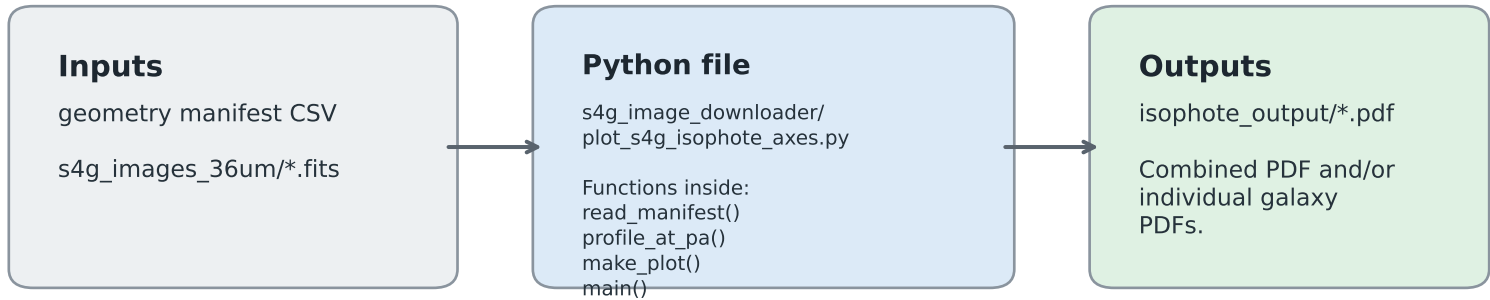
Helper files worth knowing about: `barprofile_utils.py` reads PE/VPD classifications and B/P morphology; `datautils.py` reads the Erwin table format; `plotutils.py` contains profile and frequency plotting helpers; `angle_utils.py` contains position-angle and deprojection calculations.

Workflow 1: S4G Images And Geometry



The manifest is the bridge between catalogue data and image data. Later plotting code does not need to re-query catalogues; it reads one row per galaxy from this CSV and follows the linked FITS file path.

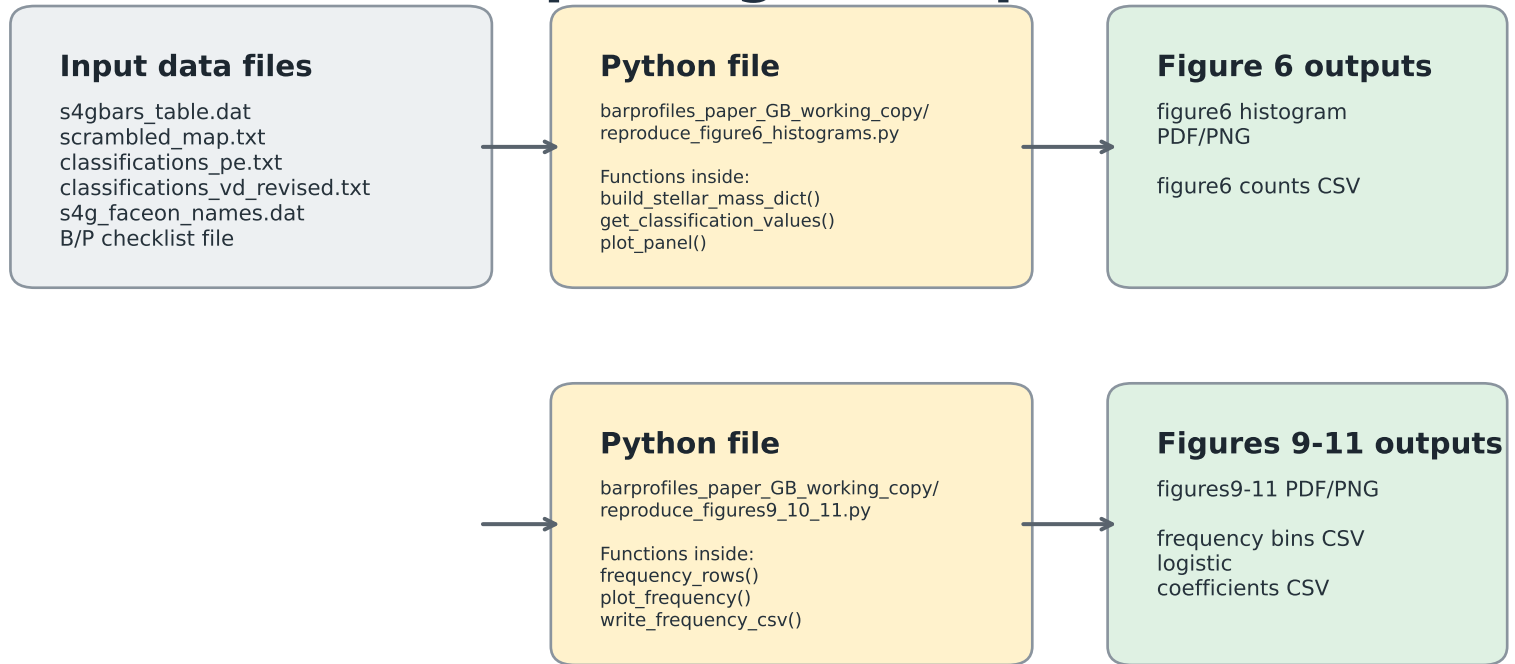
Workflow 2: Isophote And Profile Plotting



What the plotting code does

- Loads image pixels from the FITS file linked in the manifest.
- Uses catalogue geometry to place the galaxy centre, bar major axis, and projected bar minor axis.
- Draws isophote contours so the image structure can be inspected visually.
- Samples light profiles along the bar-major and bar-minor directions.
- Writes visual diagnostic PDFs; it does not create PE/VPD classifications.

Workflow 3: Paper Figure Reproduction



Important limitation

Figures 6, 9, 10, and 11 are not measured from the downloaded images alone. They depend on catalogue values such as logMstar and V_rot, plus human classifications such as PE/VPD profile classes and B/P morphology. The statistical figures reproduce the paper's tabulated inputs.